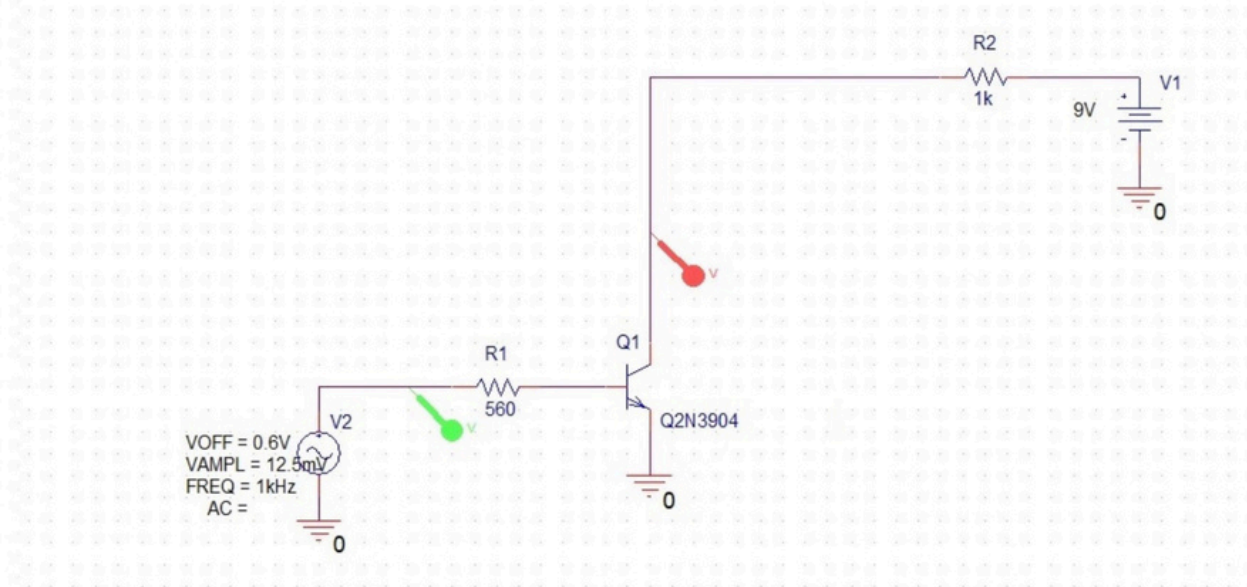
Thomas Puskas 2025952293

Experiment 7: BJT Common Emitter

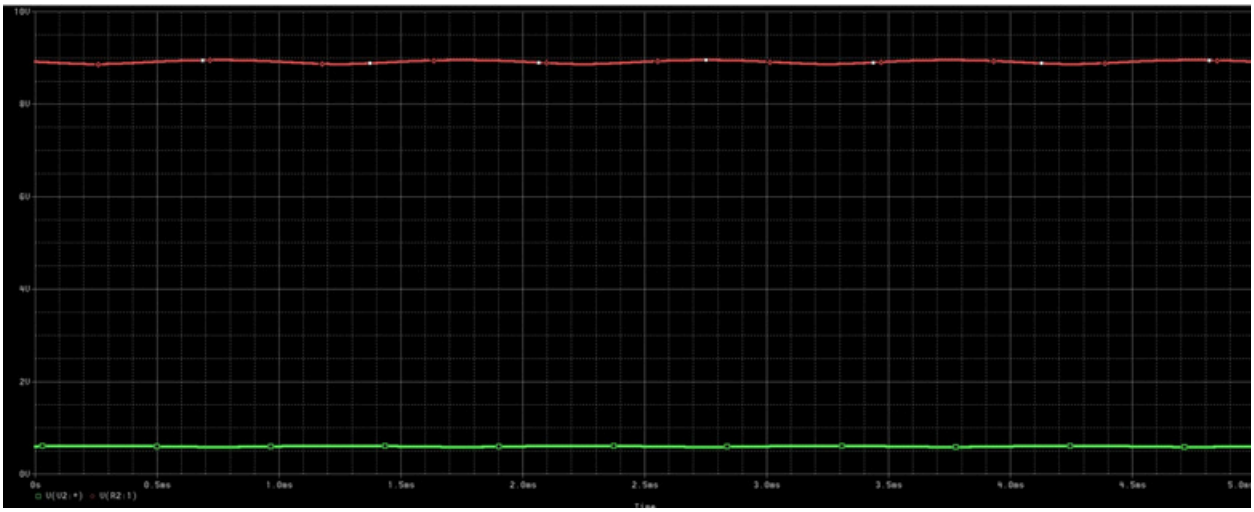
The purpose of this experiment is to see how amplifiers respond to different frequencies and see the effects of external capacitive loads and innate parasitic capacitances.

Pre-Lab:

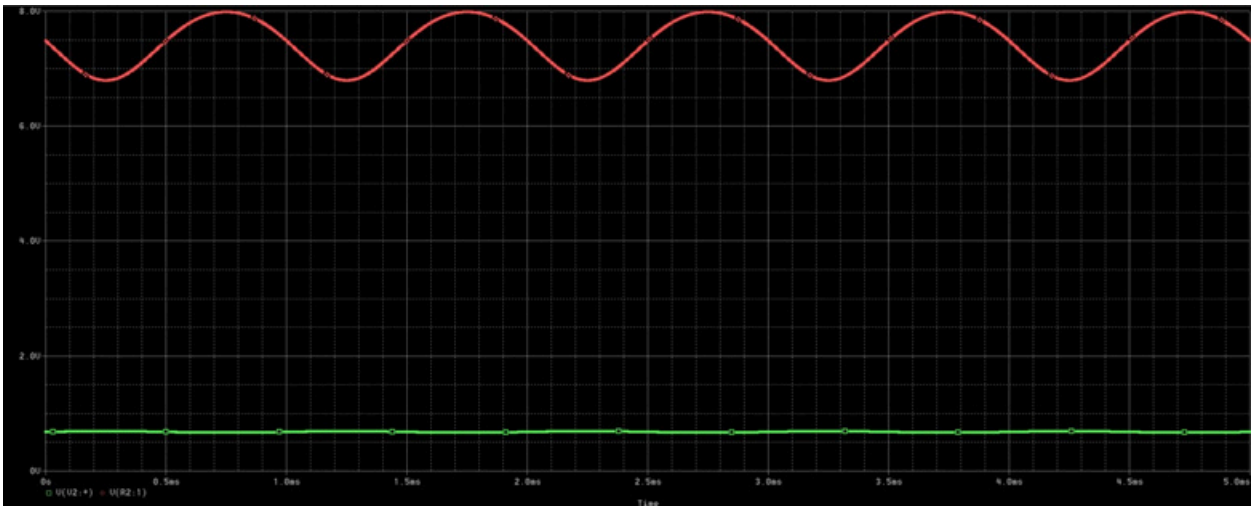
Experiment 7.1



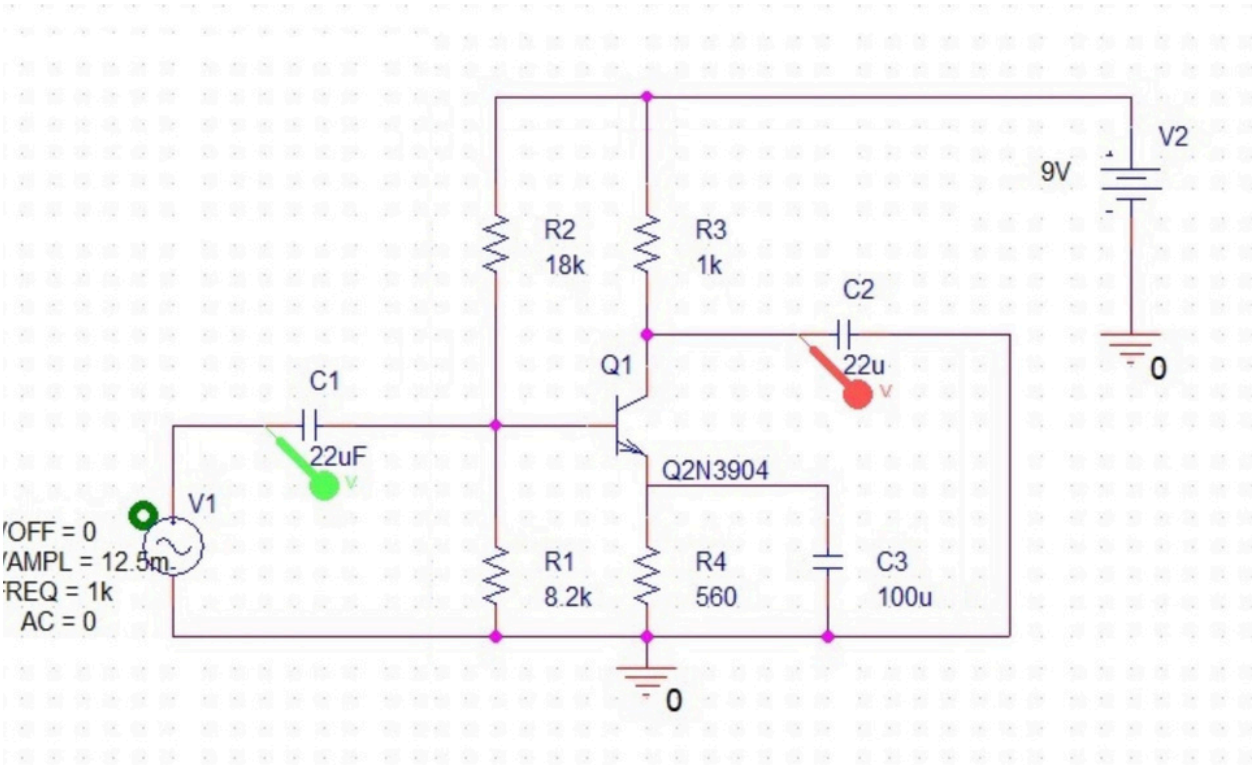
DC offset of 0.6V:

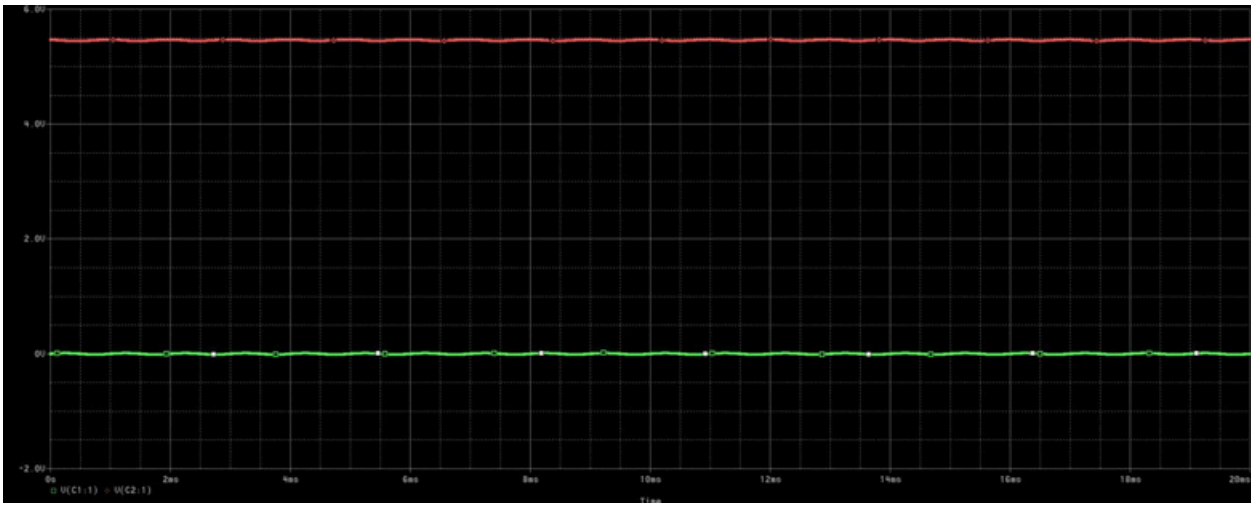


DC offset of 0.68V:

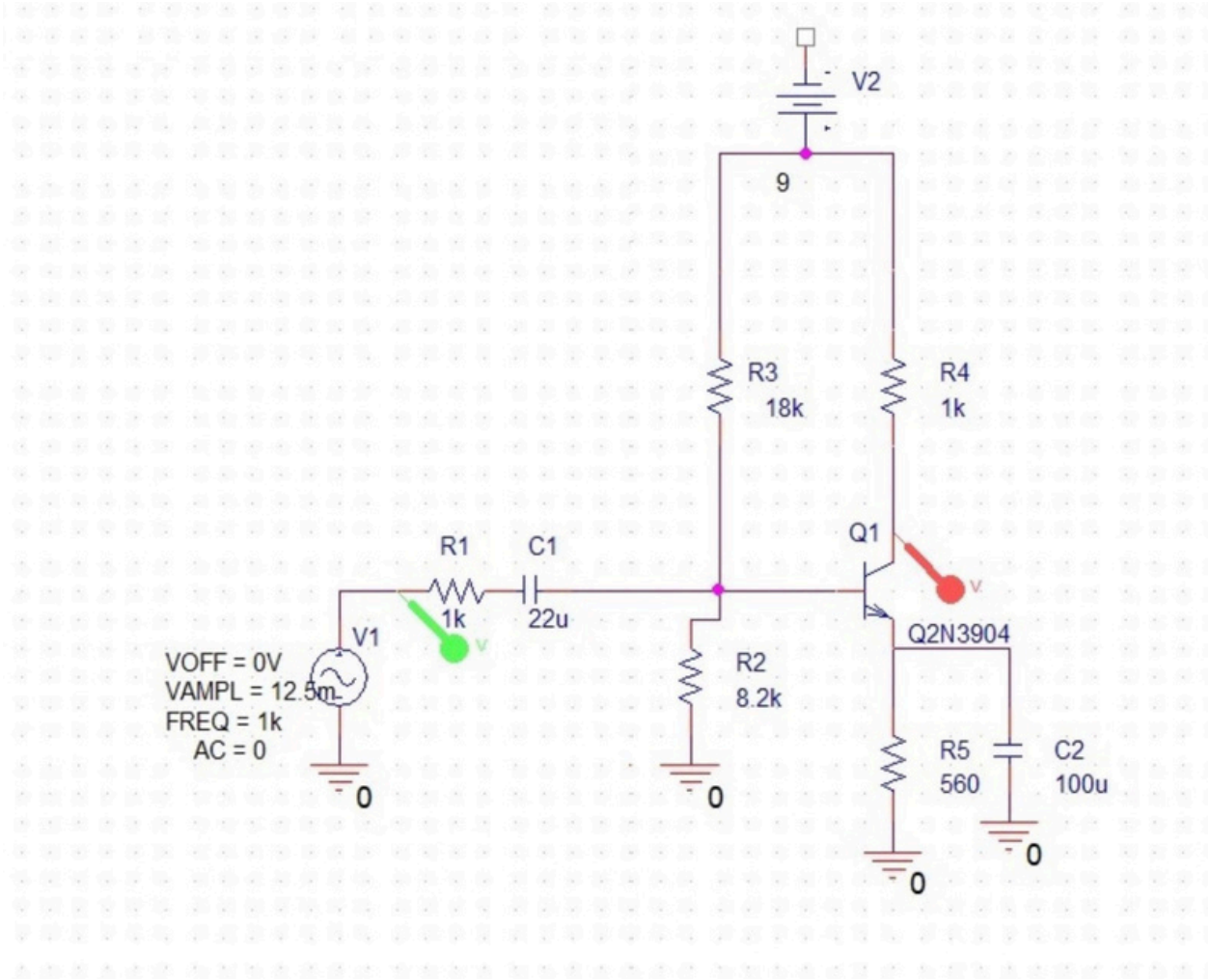


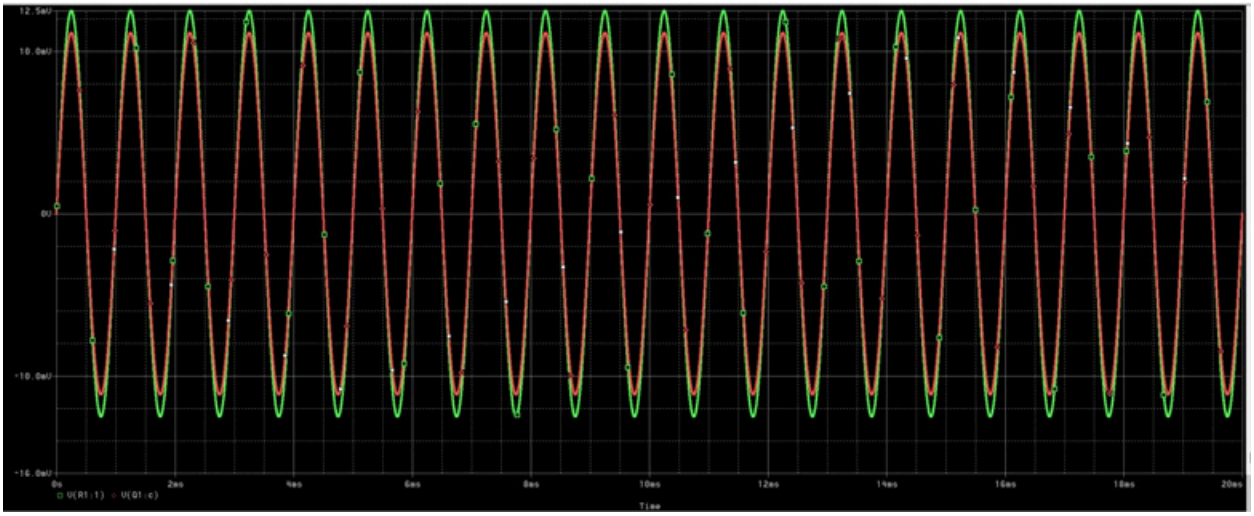
Experiment 7.2





Experiment 7.3





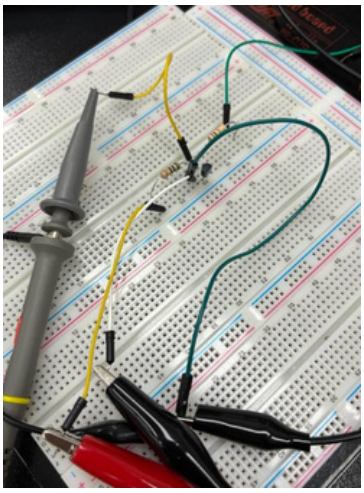
3.1 Biasing of Common Emitter

Amplifier Procedure:

- A sine wave signal with a frequency of 1 kHz and an amplitude of 25 mV peak-to-peak was generated together with a 600 mV DC offset. The combination of the AC signal and the DC bias formed the input signal, vinvin.
- The input waveform, vinvin, was connected to Channel 2 of the oscilloscope, while the output waveform, voutvout, was connected to Channel 1. The displayed waveforms were observed and captured as an image for inclusion in the report.
- The voltage gain, calculated as the ratio vout/vinvout/vin, was measured using the oscilloscope and the obtained values were entered into Table 7.1.
- The DC bias voltage was gradually varied from 600 mV to 680 mV using four different settings. For each bias value, the gain measurement procedure described in Step 3 was repeated.
- Using the collected data from Table 7.1, a graph showing the relationship between voutvout and vinvin was plotted.
- From the plotted graph, the DC bias voltage corresponding to the highest small-signal voltage gain was identified.
- Finally, an explanation was provided describing why the BJT exhibited very low gain whenever it was operated outside the forward-active region.

DC offset [mV]	Vout	Magnitude
600	5.20	173.3
630	14.2	473.3
650	25.0	833.3
680	36.8	1200

Circuit:



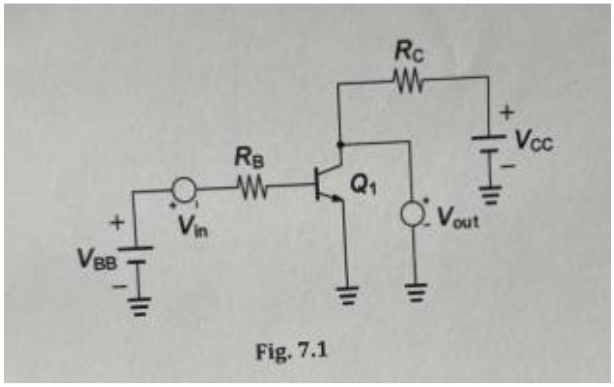
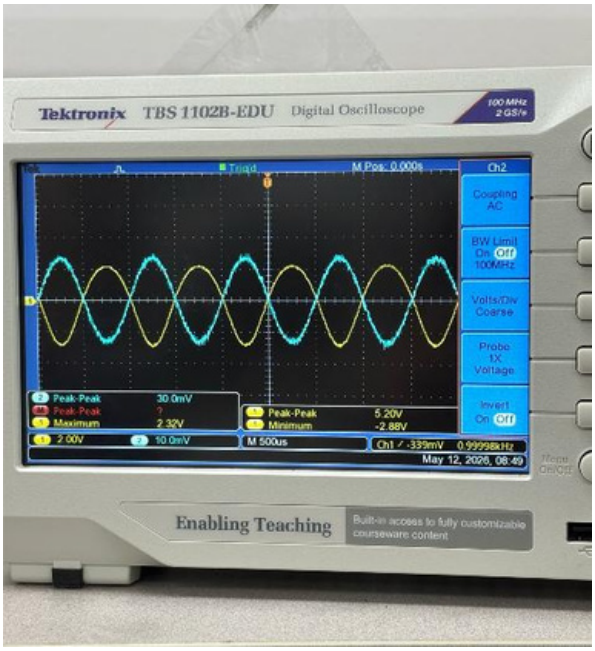
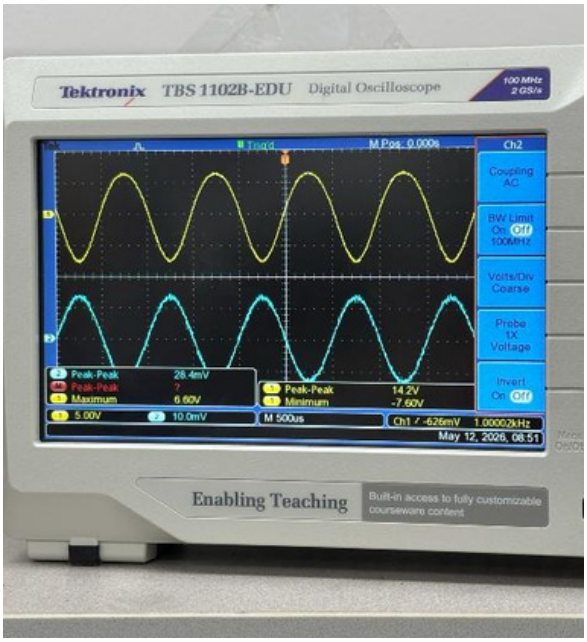


Fig. 7.1

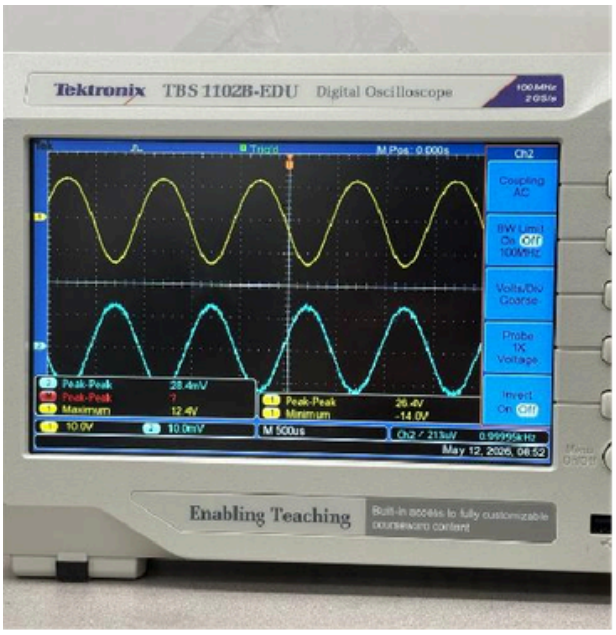
DC offset at 600mV:



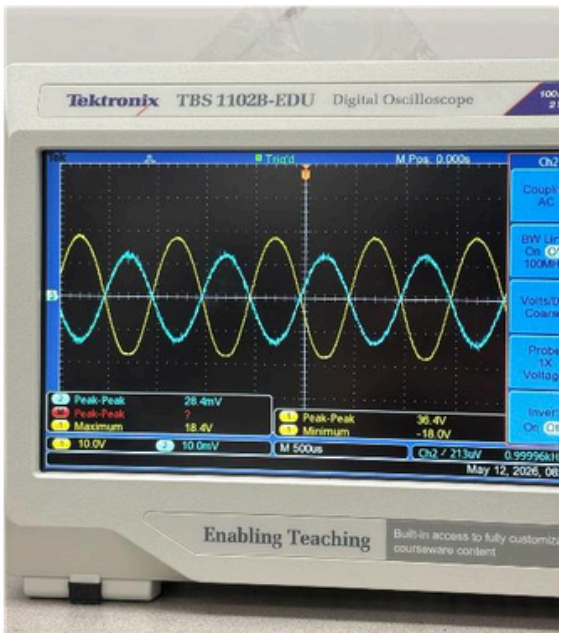
DC offset at 630mV:



DC offset at 650mV:



DC offset at 680mV:

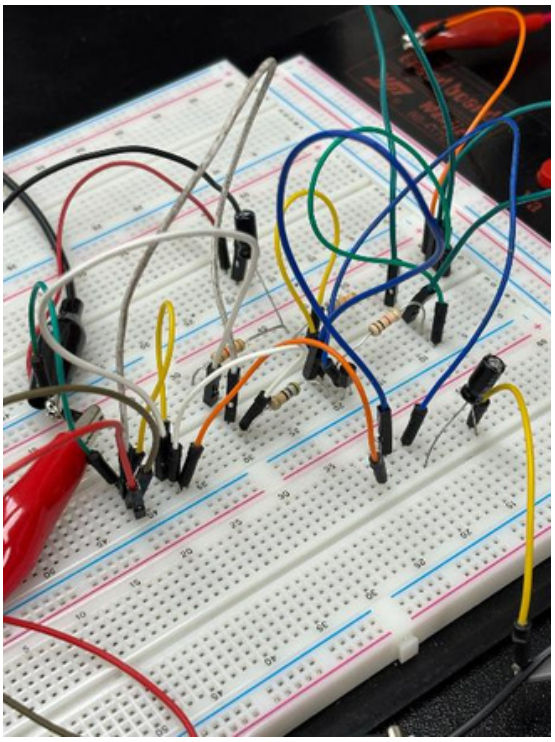


3.2 Emitter Bypass

Capacitor Procedure:

- The common emitter amplifier circuit was assembled according to the configuration illustrated in Figure 7.2.
- An oscilloscope was connected across the amplifier output terminals. A sine wave input signal with a frequency of 1 kHz, an amplitude of 25 mV peak-to-peak, and a DC offset of 0 V was applied to the circuit.
- The peak-to-peak voltages of both the input signal (measured from base to ground) and the output signal (measured from collector to ground) were observed using the oscilloscope and documented in Table 7.2. In addition, the AC waveform between the emitter and ground was measured and recorded.
- The AC voltage gain of the amplifier was determined from the measured values and entered into Table 7.2.
- Without changing the function generator settings, the bypass capacitor, CECE, was disconnected from the circuit.
- The measurements and gain calculations described in Steps 3 and 4 were repeated after removing CECE, and the new results were also recorded in Table 7.2.
- The influence of the bypass capacitor, CECE, on the overall performance and gain of the amplifier was then analyzed.

Circuit:



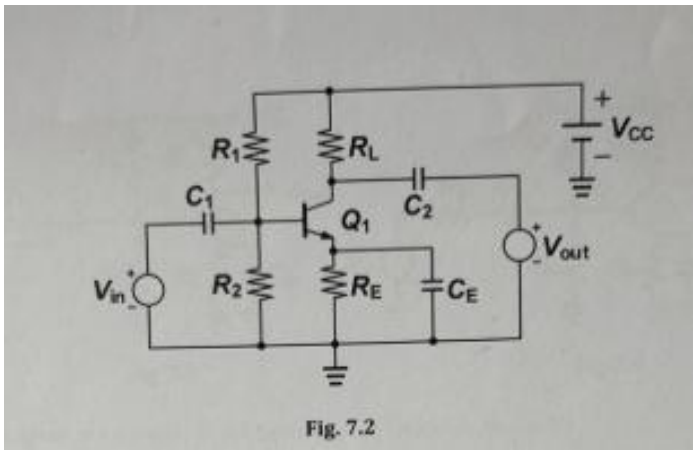
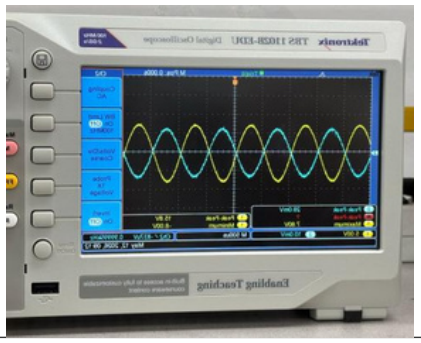
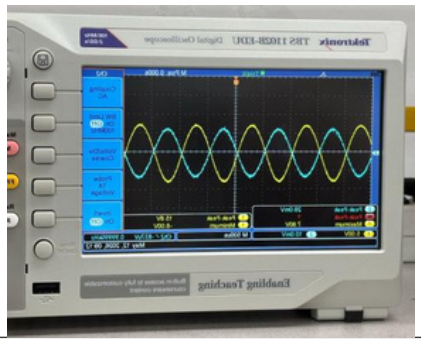
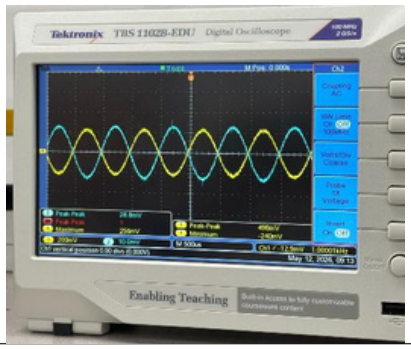
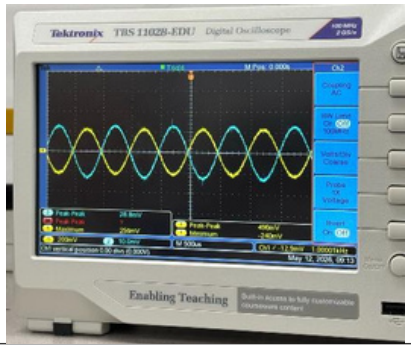


Fig. 7.2

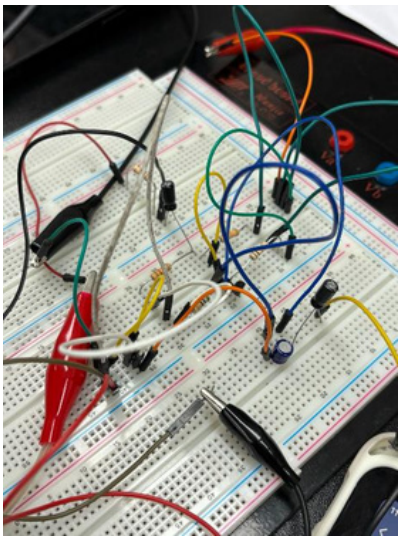
Step	Waveform		
	Base (28.8mV)	Collector (15.8)	Gain
With Ce			564.28
Without Ce	(28.4mV) 	(496mV) 	16.5

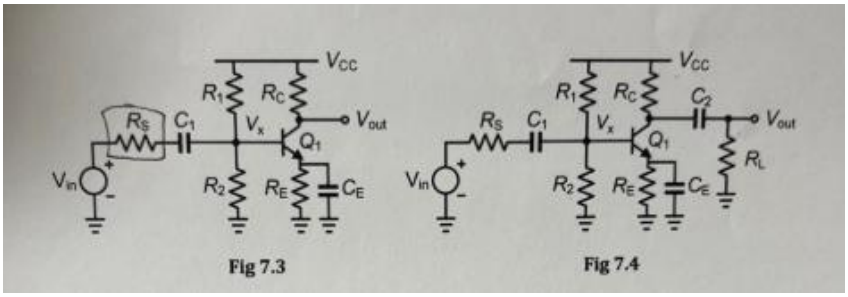
3.3 Input and Output

Impedance Procedure:

- The common emitter amplifier circuit was assembled according to the configuration shown in Figure 7.3.
- A sinusoidal input signal with a frequency of 1 kHz, an amplitude of 25 mV peak-to-peak, and a DC offset of 0 V was applied to the amplifier.
- The peak-to-peak output voltage at node VoutVout was measured using an oscilloscope and the readings were recorded in Table 7.3. The resistance values were varied according to the values specified in Table 7.3.
- The observed variation in voltage gain resulting from the change in resistance was analyzed and explained.
- Another common emitter amplifier circuit was then constructed based on the arrangement shown in Figure 7.4, where the load resistor RLRL was configured as a potentiometer.
- The peak-to-peak output voltage at node VoutVout was measured and recorded in Table 7.4. Initial measurements were taken with the load resistor RLRL disconnected from the circuit.
- Afterward, the load resistor RLRL was connected and adjusted gradually until the measured output voltage decreased to one-half of the value obtained in Step 6.
- The resistor RLRL was subsequently removed, and its resistance was measured using a multimeter and recorded in Table 7.4. This resistance value was considered to represent the output impedance, RoutRout, of the amplifier. It was also noted that the amplifier output impedance is not constant, as it varies depending on the transistor operating voltage and the connected load resistance.

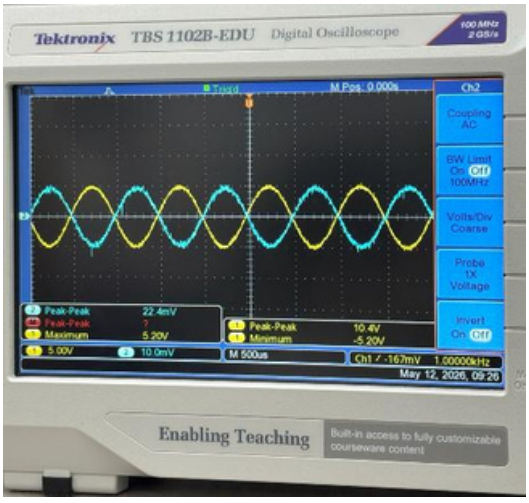
Circuit:

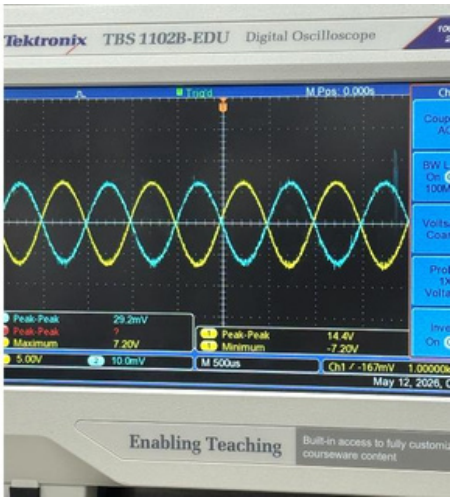


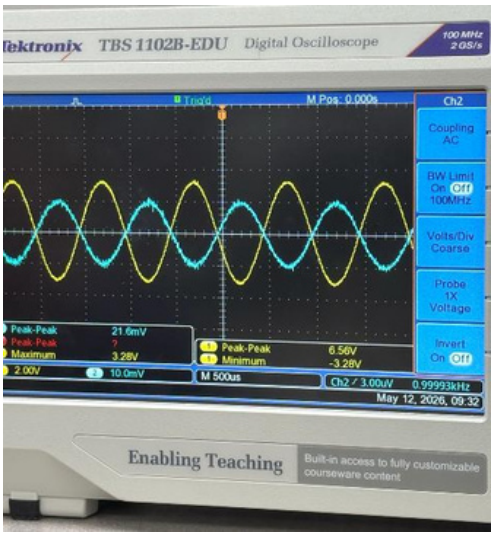


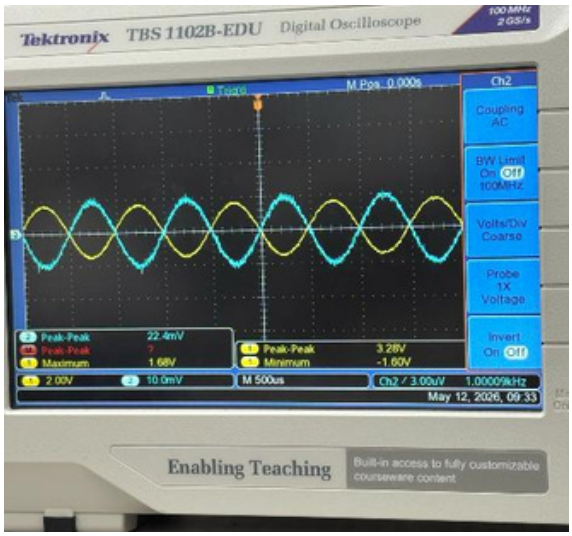
RS N/A	R1	R2	VOUT	AV
1kOh	18kOhm	8.2kOhm	14.2	467
m	18kOhm	8.2kOhm	10.6	481

RS	R1	R2	VOUT	AV	RL
1kOhm	18kOhm	8.2kOhm	10.6	481	Open
1kOhm	18kOhm	8.2kOhm	3.28	146	800Ohm
1kOhm	18kOhm	8.2kOhm	6.56	312.4	Max:4.26kOhm






Tektronix TBS 1102B-EDU Digital Oscilloscope
100 MHz 2 GS/s
Trig'd
M Pos: 0.000s
Ch2
Coupling AC
BW Limit On Off 100MHz
Volts/Div Coarse
Probe 1X Voltage
Invert On Off
Peak-Peak 21.6mV
Peak-Peak ?
Maximum 3.28V
2.00V
10.0mV
Peak-Peak 6.56V
Minimum -3.28V
M 500us
Ch2 ∕ 3.00uV 0.99993kHz
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Tektronix TBS 1102B-EDU Digital Oscilloscope
100 MHz 2 GS/s
Trig'd
M Pos: 0.000s
Ch2
Coupling AC
BW Limit On Off 100MHz
Volts/Div Coarse
Probe 1X Voltage
Invert On Off
Peak-Peak 22.4mV
Peak-Peak ?
Maximum 1.68V
2.00V
10.0mV
Peak-Peak 3.28V
Minimum -1.60V
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Enabling Teaching
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